



## Qualification Pack



# Ladle Refining Furnace - Process Control Engineer

QP Code: ISC/Q1404

Version: 1.0

NSQF Level: 6

Indian Iron and Steel Sector Skill Council || Plot no.B-7, Action Area-III, 5th Floor, Room no.509, Near Hazra Kalibari, Opposite Uni World City, New Town, Kolkata, West Bengal  
Pin-700156 || email:akash@iisssc.org



## Qualification Pack

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## Qualification Pack

# ISC/Q1404: Ladle Refining Furnace - Process Control Engineer

## Brief Job Description

The individual in this role coordinate and monitor the Ladle Refining Furnace operation and its associated functions.

## Personal Attributes

The person should be patient, organized, team-oriented and have the ability to work for long hours in adverse conditions. They must be able to plan and prioritise tasks effectively and have an eye for detail and quality.

## Applicable National Occupational Standards (NOS)

### Compulsory NOS:

1. [ISC/N0946: Manage and implement health and safety practices at the work place](#)
2. [ISC/N0947: Manage team and others at workplace](#)
3. [ISC/N1407: Prepare for Ladle Refining Furnace \(LRF\) Process](#)
4. [ISC/N1412: Perform and control Ladle Refining Furnace\(LRF\) process](#)
5. [ISC/N1704: Apply Decarbonization practice in plant](#)
6. [ISC/N1703: Apply statistical process control \(SPC\) techniques](#)
7. [DGT/VSQ/N0103: Employability Skills \(90 Hours\)](#)

## Qualification Pack (QP) Parameters

|                   |                |
|-------------------|----------------|
| <b>Sector</b>     | Iron and Steel |
| <b>Sub-Sector</b> | Steel          |
| <b>Occupation</b> | Steel Making   |
| <b>Country</b>    | India          |
| <b>NSQF Level</b> | 6              |
| <b>Credits</b>    | 20             |



## Qualification Pack

|                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-----------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Aligned to NCO/ISCO/ISIC Code</b>                      | NCO-2015/3122.0100                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Minimum Educational Qualification &amp; Experience</b> | Completed 4 year UG program (BE/ B. Tech in Metallurgy / Material Science / Mechanical / Chemical/ Production)<br>OR<br>Completed 2nd year diploma after 12th (in Metallurgy / Material Science / Mechanical / Chemical/ Production) with 3 Years of experience<br>OR<br>Completed 3 year diploma after 10th (in Metallurgy / Material Science / Mechanical / Chemical/ Production) with 3-5 Years of experience 4.5 Years<br>OR<br>Previous relevant Qualification of NSQF Level (5) with 3 Years of experience |
| <b>Minimum Level of Education for Training in School</b>  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Pre-Requisite License or Training</b>                  | NA                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Minimum Job Entry Age</b>                              | 20 Years                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Last Reviewed On</b>                                   | NA                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Next Review Date</b>                                   | 30/04/2027                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>NSQC Approval Date</b>                                 | 30/04/2024                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Version</b>                                            | 1.0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Reference code on NQR</b>                              | QG-06-IS-02480-2024-V1-IISSSC                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>NQR Version</b>                                        | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |



## Qualification Pack

# ISC/N0946: Manage and implement health and safety practices at the work place

## Description

This OS unit is about managing and implementing health, hygiene, safety and sustainable practices for optimising use of resources.

## Scope

The scope covers the following :

- Maintain safe and secure working environment
- Maintain health and hygiene
- Manage housekeeping at workplace
- Implement material and energy conservation practices

## Elements and Performance Criteria

### *Maintain safe and secure working environment*

To be competent, the user/individual on the job must be able to:

- PC1.** identify hazardous activities and the possible causes of risks or accidents in the workplace
- PC2.** implement safe working practices while dealing with hazards to ensure safety of self and others
- PC3.** ensure that team is following appropriate safety practices while working in and around trenches, elevated places and confined areas
- PC4.** conduct regular checks of the machines with support of the maintenance team to identify potential hazards
- PC5.** organise safety drills or training sessions to create awareness amongst others on the identified risks and safety practices
- PC6.** fill daily check sheet to report improvements done and risks identified
- PC7.** ensure that relevant safety boards/signs are placed on the shop floor for the safety of self and others
- PC8.** report any identified breaches in health, safety and security policies and procedures to the designated person
- PC9.** ensure that team is following appropriate emergency procedures such as raising alarm, safe evacuation etc.

### *Maintain health and hygiene*

To be competent, the user/individual on the job must be able to:

- PC10.** ensure team is aware about hygiene and sanitation regulations and following them on the shop floor
- PC11.** ensure workplace, equipment, restrooms etc. are cleaned and sanitized regularly
- PC12.** ensure availability of running water, hand wash and alcohol-based sanitizers at the workplace
- PC13.** report advanced hygiene and sanitation issues to appropriate authority



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**PC14.** report regarding the contagious illness of self or people in close contact

### *Manage housekeeping at workplace*

To be competent, the user/individual on the job must be able to:

**PC15.** implement appropriate housekeeping methods and practices at workplace

**PC16.** ensure good housekeeping in order to prevent hazards and accidents

**PC17.** monitor that that team is storing the material, tools and equipment in the correct location and in good condition after work completion

**PC18.** ensure recyclable, non-recyclable and hazardous wastes are segregated as per SOP

**PC19.** ensure proper mechanism is followed while collecting and disposing of non-recyclable, recyclable and reusable waste

### *Implement material and energy conservation practices*

To be competent, the user/individual on the job must be able to:

**PC20.** identify processes where material and energy/electricity utilization can be optimized

**PC21.** ensure malfunctioning (fumes/sparks/emission/vibration/noise) and lapse in maintenance of equipment are resolved effectively

**PC22.** prepare and analyze material and energy audit reports to decipher excessive consumption of material and water

**PC23.** identify possibilities of using renewable energy and environment friendly fuels

## Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

**KU1.** organisation procedures for health, safety and security, individual role and responsibilities in this context

**KU2.** the organisation's emergency procedures for different emergency situations and the importance of following the same

**KU3.** evacuation procedures for workers and visitors

**KU4.** how and when to report hazards

**KU5.** potential hazards, risks and threats based on the nature of work

**KU6.** preventative and remedial actions to be taken in case of exposure to toxic material

**KU7.** various types of fire extinguisher

**KU8.** various types of safety signs and their meaning

**KU9.** appropriate first aid treatment relevant to different condition e.g. bleeding, minor burns, eye injuries etc.

**KU10.** relevant standards, procedures and policies related to 5S followed in the company

**KU11.** the various materials used and their storage norms

**KU12.** efficient utilisation of material and water

**KU13.** basics of electricity and prevalent energy efficient devices

**KU14.** common practices of conserving electricity

**KU15.** categorisation of waste into dry, wet, recyclable, non-recyclable and items of single-use plastics

**KU16.** usage of different colors of dustbins



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**KU17.** waste management techniques

### Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read safety instructions/guidelines
- GS2.** modify work practices to improve them
- GS3.** work with supervisors/team members to carry out work related tasks
- GS4.** complete tasks efficiently and accurately within stipulated time
- GS5.** inform/report to concerned person in case of any problem
- GS6.** make timely decisions for efficient utilization of resources
- GS7.** write reports such as accident report, in at least English/regional language



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### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                                | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|-------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Maintain safe and secure working environment</i>                                                                                             | <b>12</b>    | <b>21</b>       | -             | <b>8</b>   |
| <b>PC1.</b> identify hazardous activities and the possible causes of risks or accidents in the workplace                                        | 2            | 3               | -             | 1          |
| <b>PC2.</b> implement safe working practices while dealing with hazards to ensure safety of self and others                                     | 2            | 2               | -             | 1          |
| <b>PC3.</b> ensure that team is following appropriate safety practices while working in and around trenches, elevated places and confined areas | 1            | 2               | -             | -          |
| <b>PC4.</b> conduct regular checks of the machines with support of the maintenance team to identify potential hazards                           | 2            | 4               | -             | 1          |
| <b>PC5.</b> organise safety drills or training sessions to create awareness amongst others on the identified risks and safety practices         | 1            | 2               | -             | 1          |
| <b>PC6.</b> fill daily check sheet to report improvements done and risks identified                                                             | 1            | 2               | -             | 1          |
| <b>PC7.</b> ensure that relevant safety boards/signs are placed on the shop floor for the safety of self and others                             | 1            | 2               | -             | 1          |
| <b>PC8.</b> report any identified breaches in health, safety and security policies and procedures to the designated person                      | 1            | 2               | -             | 1          |
| <b>PC9.</b> ensure that team is following appropriate emergency procedures such as raising alarm, safe evacuation etc.                          | 1            | 2               | -             | 1          |
| <i>Maintain health and hygiene</i>                                                                                                              | <b>6</b>     | <b>9</b>        | -             | <b>4</b>   |
| <b>PC10.</b> ensure team is aware about hygiene and sanitation regulations and following them on the shop floor                                 | 2            | 3               | -             | 1          |
| <b>PC11.</b> ensure workplace, equipment, restrooms etc. are cleaned and sanitized regularly                                                    | 1            | 2               | -             | 1          |





## Qualification Pack

| Assessment Criteria for Outcomes                                                                                                                     | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC12.</b> ensure availability of running water, hand wash and alcohol-based sanitizers at the workplace                                           | 1            | 2               | -             | 1          |
| <b>PC13.</b> report advanced hygiene and sanitation issues to appropriate authority                                                                  | 1            | 1               | -             | 1          |
| <b>PC14.</b> report regarding the contagious illness of self or people in close contact                                                              | 1            | 1               | -             | -          |
| <i>Manage housekeeping at workplace</i>                                                                                                              | <b>6</b>     | <b>10</b>       | -             | <b>4</b>   |
| <b>PC15.</b> implement appropriate housekeeping methods and practices at workplace                                                                   | 2            | 3               | -             | 1          |
| <b>PC16.</b> ensure good housekeeping in order to prevent hazards and accidents                                                                      | 1            | 2               | -             | 1          |
| <b>PC17.</b> monitor that that team is storing the material, tools and equipment in the correct location and in good condition after work completion | 1            | 2               | -             | 1          |
| <b>PC18.</b> ensure recyclable, non-recyclable and hazardous wastes are segregated as per SOP                                                        | 1            | 2               | -             | 1          |
| <b>PC19.</b> ensure proper mechanism is followed while collecting and disposing of non-recyclable, recyclable and reusable waste                     | 1            | 1               | -             | -          |
| <i>Implement material and energy conservation practices</i>                                                                                          | <b>6</b>     | <b>10</b>       | -             | <b>4</b>   |
| <b>PC20.</b> identify processes where material and energy/electricity utilization can be optimized                                                   | 2            | 3               | -             | 1          |
| <b>PC21.</b> ensure malfunctioning (fumes/sparks/emission/vibration/noise) and lapse in maintenance of equipment are resolved effectively            | 1            | 2               | -             | 1          |
| <b>PC22.</b> prepare and analyze material and energy audit reports to decipher excessive consumption of material and water                           | 2            | 3               | -             | 1          |
| <b>PC23.</b> identify possibilities of using renewable energy and environment friendly fuels                                                         | 1            | 2               | -             | 1          |
| <b>NOS Total</b>                                                                                                                                     | <b>30</b>    | <b>50</b>       | -             | <b>20</b>  |



## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                                                                                                                                                                  |
|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>NOS Code</b>            | ISC/N0946                                                                                                                                                                        |
| <b>NOS Name</b>            | Manage and implement health and safety practices at the work place                                                                                                               |
| <b>Sector</b>              | Iron and Steel                                                                                                                                                                   |
| <b>Sub-Sector</b>          | Steel, Sponge Iron, Ferro Alloys, Re-Rollers, Refractory                                                                                                                         |
| <b>Occupation</b>          | Mechanical Maintenance, Raw Material Handling , Product Quality Control, Refractory Maintenance, Furnace Operation, Fitter, Cold Rolling Mill Operations, Electrical Maintenance |
| <b>NSQF Level</b>          | 5                                                                                                                                                                                |
| <b>Credits</b>             | 2                                                                                                                                                                                |
| <b>Version</b>             | 1.0                                                                                                                                                                              |
| <b>Last Reviewed Date</b>  | 07/10/2025                                                                                                                                                                       |
| <b>Next Review Date</b>    | 07/10/2028                                                                                                                                                                       |
| <b>NSQC Clearance Date</b> | 07/10/2025                                                                                                                                                                       |



## Qualification Pack

### ISC/N0947: Manage team and others at workplace

#### Description

This OS unit is about managing team, colleagues/superiors and others, either in own work group or in other work groups within organisation.

#### Scope

The scope covers the following :

- Communicate effectively with team members
- Interact with supervisor
- Manage appropriate behaviour at work place

#### Elements and Performance Criteria

##### *Communicate effectively with team members*

To be competent, the user/individual on the job must be able to:

- PC1.** manage and co-ordinate with team members to integrate work, as per the workload in order to achieve team goals
- PC2.** implement ways to share information with team members in line with organisational requirements
- PC3.** ensure that work requirements are clearly communicated to the team members through all means including face-to-face, telephonic and written
- PC4.** work in a way that show respect for all team members and customers
- PC5.** carry out commitments made to team members
- PC6.** resolve conflicts within the team members at work to achieve smooth workflow
- PC7.** guide the team members to follow the organisation's policies and procedures
- PC8.** ensure team goals are given preference over individual goals
- PC9.** encourage team members to rectify errors as per feedback and minimize mistakes in future

##### *Interact with supervisor*

To be competent, the user/individual on the job must be able to:

- PC10.** report progress on job allocated and team performance to the superiors
- PC11.** escalate problems to supervisors that cannot be handled
- PC12.** report the completed work
- PC13.** train the team members to report completed work and receive feedback on work done

##### *Manage appropriate behaviour at work place*

To be competent, the user/individual on the job must be able to:

- PC14.** help PwD team members to overcome the challenges, if asked
- PC15.** ensure team shows sensitivity towards all genders and PwD
- PC16.** adjust communication styles to reflect gender sensitivity and sensitivity towards person with disability



## Qualification Pack

**PC17.** ensure that team is following responsible and disciplined behaviours at the workplace

### Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the importance of effective communication and establishing good working relationships with colleagues and supervisor
- KU2.** different methods of communication as per the circumstances
- KU3.** importance of teamwork in organization and individual success
- KU4.** various components of effective communication
- KU5.** barriers to effective communication
- KU6.** common reasons for interpersonal conflict
- KU7.** what constitutes disciplined behaviour for a working professional
- KU8.** gender concepts, issues & legislation
- KU9.** organisational policies and procedures related to gender equality
- KU10.** challenges faced by PWD and the ways to help them overcome the same

### Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read instructions/guidelines/procedures
- GS2.** listen effectively and orally communicate information
- GS3.** ask for clarification and advice from the concerned person
- GS4.** maintain positive and effective relationships with colleagues
- GS5.** evaluate the possible solution(s) to the problem
- GS6.** spot and communicate potential areas of disruptions in the work process and report the same
- GS7.** complete written work with attention to detail
- GS8.** check that the work meets customer requirements



## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                                        | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|---------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Communicate effectively with team members</i>                                                                                                        | <b>14</b>    | <b>24</b>       | -             | <b>10</b>  |
| <b>PC1.</b> manage and co-ordinate with team members to integrate work, as per the workload in order to achieve team goals                              | 3            | 5               | -             | 2          |
| <b>PC2.</b> implement ways to share information with team members in line with organisational requirements                                              | 3            | 5               | -             | 2          |
| <b>PC3.</b> ensure that work requirements are clearly communicated to the team members through all means including face-to-face, telephonic and written | 2            | 4               | -             | 2          |
| <b>PC4.</b> work in a way that show respect for all team members and customers                                                                          | 1            | 2               | -             | 1          |
| <b>PC5.</b> carry out commitments made to team members                                                                                                  | 1            | 1               | -             | 1          |
| <b>PC6.</b> resolve conflicts within the team members at work to achieve smooth workflow                                                                | 1            | 2               | -             | -          |
| <b>PC7.</b> guide the team members to follow the organisation's policies and procedures                                                                 | 1            | 2               | -             | 1          |
| <b>PC8.</b> ensure team goals are given preference over individual goals                                                                                | 1            | 1               | -             | 1          |
| <b>PC9.</b> encourage team members to rectify errors as per feedback and minimize mistakes in future                                                    | 1            | 2               | -             | -          |
| <i>Interact with supervisor</i>                                                                                                                         | <b>8</b>     | <b>10</b>       | -             | <b>6</b>   |
| <b>PC10.</b> report progress on job allocated and team performance to the superiors                                                                     | 2            | 2               | -             | 1          |
| <b>PC11.</b> escalate problems to supervisors that cannot be handled                                                                                    | 2            | 3               | -             | 2          |
| <b>PC12.</b> report the completed work                                                                                                                  | 2            | 2               | -             | 1          |



## Qualification Pack

| Assessment Criteria for Outcomes                                                                                      | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|-----------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC13.</b> train the team members to report completed work and receive feedback on work done                        | 2            | 3               | -             | 2          |
| <i>Manage appropriate behaviour at work place</i>                                                                     | <b>8</b>     | <b>16</b>       | -             | <b>4</b>   |
| <b>PC14.</b> help PwD team members to overcome the challenges, if asked                                               | 2            | 4               | -             | 1          |
| <b>PC15.</b> ensure team shows sensitivity towards all genders and PwD                                                | 2            | 4               | -             | 1          |
| <b>PC16.</b> adjust communication styles to reflect gender sensitivity and sensitivity towards person with disability | 2            | 4               | -             | 1          |
| <b>PC17.</b> ensure that team is following responsible and disciplined behaviours at the workplace                    | 2            | 4               | -             | 1          |
| <b>NOS Total</b>                                                                                                      | <b>30</b>    | <b>50</b>       | -             | <b>20</b>  |



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### National Occupational Standards (NOS) Parameters

|                            |                                                                                                                                                                                                                                                                                                             |
|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>NOS Code</b>            | ISC/N0947                                                                                                                                                                                                                                                                                                   |
| <b>NOS Name</b>            | Manage team and others at workplace                                                                                                                                                                                                                                                                         |
| <b>Sector</b>              | Iron and Steel                                                                                                                                                                                                                                                                                              |
| <b>Sub-Sector</b>          | Steel, Sponge Iron, Ferro Alloys, Re-Rollers, Refractory, Mechanical Maintenance, Iron Making                                                                                                                                                                                                               |
| <b>Occupation</b>          | Mechanical Maintenance, Raw Material Handling , Coke Making, Operations, Iron Making, Cold Rolling Mill Operations, Product Quality Control, Electrical Maintenance, Electronics and Instrumentation Maintenance, Refractory Maintenance, Furnace Operation, Fitter, Steel Making, Pre-fabricated Structure |
| <b>NSQF Level</b>          | 5                                                                                                                                                                                                                                                                                                           |
| <b>Credits</b>             | 1                                                                                                                                                                                                                                                                                                           |
| <b>Version</b>             | 1.0                                                                                                                                                                                                                                                                                                         |
| <b>Last Reviewed Date</b>  | 07/10/2025                                                                                                                                                                                                                                                                                                  |
| <b>Next Review Date</b>    | 07/10/2028                                                                                                                                                                                                                                                                                                  |
| <b>NSQC Clearance Date</b> | 07/10/2025                                                                                                                                                                                                                                                                                                  |



## Qualification Pack

# ISC/N1407: Prepare for Ladle Refining Furnace (LRF) Process

## Description

This OS unit is about performing preparatory activities for carrying out various Ladle Refining Furnace operations

## Scope

The scope covers the following :

- Carry out preparatory activities

## Elements and Performance Criteria

### *Carry out preparatory activities*

To be competent, the user/individual on the job must be able to:

- PC1.** obtain the required information and work completed in previous shift from logbook, discussions etc. and take handover from the previous shift operator by following organizational procedures
- PC2.** read work orders, blueprints etc. to determine setup specifications, work sequences, grade of steel required
- PC3.** identify the type of ladle i.e. new ladle or circulation ladle available for the secondary steel processing
- PC4.** clean the ladle by using appropriate cleaning agents and following SOP incase of circulation ladle
- PC5.** heat the ladle at required temperature or as per the specified norms
- PC6.** use appropriate Personal Protective Equipment (PPE) for safe working in work area
- PC7.** ensure availability of sufficient quantity of primary liquid steel, and other material like deoxidisers, additives (C, Mn, Si, Al), flux (lime), oxygen and argon gas for desulphurization required for smooth operations and achieving production target
- PC8.** fill the liquid steel in the ladle from primary steel making sections such as EAF/BOF/IF by following organizational procedures
- PC9.** check chemical composition, temperature and dissolved O<sub>2</sub> (in ppm) of the liquid steel by following SOP
- PC10.** move the ladle to the LRF section using rollers and place it on the ladle platform properly
- PC11.** ensure that work area is clean and free of any obstacle or encumbrance
- PC12.** ensure that all safety measures, interlocks, alarms are working and in conformance with the safe operation of ladle refining furnace as per SOP
- PC13.** coordinate with other related departments such as raw material handling and quality departments

## Knowledge and Understanding (KU)

The individual on the job needs to know and understand:





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- KU1.** relevant standards and procedures followed in the company
- KU2.** types of ladle furnaces
- KU3.** Standard Operating Procedures (SOP) recommended by OEM for using tools and equipment required during LRF process
- KU4.** basic working process of ladle refining furnace
- KU5.** material required for LRF process
- KU6.** SOP as recommended by competent authority of the organisation
- KU7.** normal running characteristics of LRF equipment
- KU8.** impact of parameters i.e temperature, O<sub>2</sub>, chemical composition etc. on the final product
- KU9.** material like deoxidisers, additives (C, Mn, Si, Al), flux (lime) required during LRF process
- KU10.** need of oxygen and argon gas during LRF process
- KU11.** emergency procedures and safety measures such as gas safety, hot metal handling safety etc. associated with LRF process

## Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read work instructions, equipment manuals and process documents
- GS2.** communicate the process requirements to the supervisor and co-workers
- GS3.** attentively listen and comprehend the information given by the supervisor/team members
- GS4.** write work related information in English/regional language
- GS5.** recognize a workplace problem and take suitable action
- GS6.** plan and organize work according to the work requirements
- GS7.** complete the assigned tasks with minimum supervision
- GS8.** report to the supervisor or deal with a colleague individually, depending on the type of concern



## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                                                                                                                                                      | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Carry out preparatory activities</i>                                                                                                                                                                                                                               | 30           | 50              | -             | 20         |
| <b>PC1.</b> obtain the required information and work completed in previous shift from logbook, discussions etc. and take handover from the previous shift operator by following organizational procedures                                                             | 2            | 3               | -             | 1          |
| <b>PC2.</b> read work orders, blueprints etc. to determine setup specifications, work sequences, grade of steel required                                                                                                                                              | 2            | 3               | -             | 2          |
| <b>PC3.</b> identify the type of ladle i.e. new ladle or circulation ladle available for the secondary steel processing                                                                                                                                               | 2            | 3               | -             | 1          |
| <b>PC4.</b> clean the ladle by using appropriate cleaning agents and following SOP incase of circulation ladle                                                                                                                                                        | 2            | 3               | -             | 1          |
| <b>PC5.</b> heat the ladle at required temperature or as per the specified norms                                                                                                                                                                                      | 2            | 3               | -             | 2          |
| <b>PC6.</b> use appropriate Personal Protective Equipment (PPE) for safe working in work area                                                                                                                                                                         | -            | 1               | -             | -          |
| <b>PC7.</b> ensure availability of sufficient quantity of primary liquid steel, and other material like deoxidisers, additives (C, Mn, Si, Al), flux (lime), oxygen and argon gas for desulphurization required for smooth operations and achieving production target | 4            | 7               | -             | 3          |
| <b>PC8.</b> fill the liquid steel in the ladle from primary steel making sections such as EAF/BOF/IF by following organizational procedures                                                                                                                           | 4            | 7               | -             | 3          |
| <b>PC9.</b> check chemical composition, temperature and dissolved O <sub>2</sub> (in ppm) of the liquid steel by following SOP                                                                                                                                        | 4            | 7               | -             | 3          |
| <b>PC10.</b> move the ladle to the LRF section using rollers and place it on the ladle platform properly                                                                                                                                                              | 3            | 5               | -             | 2          |



## Qualification Pack

| Assessment Criteria for Outcomes                                                                                                                             | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC11.</b> ensure that work area is clean and free of any obstacle or encumbrance                                                                          | 2            | 3               | -             | 1          |
| <b>PC12.</b> ensure that all safety measures, interlocks, alarms are working and in conformance with the safe operation of ladle refining furnace as per SOP | 2            | 3               | -             | 1          |
| <b>PC13.</b> coordinate with other related departments such as raw material handling and quality departments                                                 | 1            | 2               | -             | -          |
| <b>NOS Total</b>                                                                                                                                             | <b>30</b>    | <b>50</b>       | <b>-</b>      | <b>20</b>  |



## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                                  |
|----------------------------|--------------------------------------------------|
| <b>NOS Code</b>            | ISC/N1407                                        |
| <b>NOS Name</b>            | Prepare for Ladle Refining Furnace (LRF) Process |
| <b>Sector</b>              | Iron and Steel                                   |
| <b>Sub-Sector</b>          |                                                  |
| <b>Occupation</b>          | Steel Making, Steel Making                       |
| <b>NSQF Level</b>          | 6                                                |
| <b>Credits</b>             | 1                                                |
| <b>Version</b>             | 1.0                                              |
| <b>Last Reviewed Date</b>  | 30/04/2024                                       |
| <b>Next Review Date</b>    | 30/04/2027                                       |
| <b>NSQC Clearance Date</b> | 30/04/2024                                       |



## Qualification Pack

# ISC/N1412: Perform and control Ladle Refining Furnace(LRF) process

## Description

This OS unit is about performing various Ladle Refining Furnace operations as per the final output specifications and the standards specified by the organization

## Scope

The scope covers the following :

- Perform LRF operations
- Perform post-operation activities

## Elements and Performance Criteria

### *Perform LRF process*

To be competent, the user/individual on the job must be able to:

- PC1.** plan the strategy and sequence for adding deoxidizers (i.e. type and quantity), additives and flux in the liquid steel on the basis of steel grade and specification required
- PC2.** add deoxidizers (i.e. type and quantity), additives and flux in the liquid steel as per the planned sequence
- PC3.** control the flow of O<sub>2</sub> in the furnace as per the requirement
- PC4.** heat the liquid steel by using electrodes available in the furnace
- PC5.** maintain the voltage, current and power in the electrodes as per the requirement
- PC6.** carry out desulphurization process for removing the Sulphur content in the liquid steel by adding more O<sub>2</sub>, lime etc. in it
- PC7.** carry out argon purging in the furnace by opening the nozzle below the ladle during the desulphurization process
- PC8.** maintain the flow of, pressure and effectiveness of argon gas during purging
- PC9.** open the tap, at the bottom of ladle and collect the sample of steel to check the Sulphur quantity as per the customer requirement
- PC10.** maintain the temperature of steel as per the requirement
- PC11.** add the alloys (ferro alloys) in the steel as per the customer requirement
- PC12.** clean the steel and remove the inclusions by “Ca” treatment process
- PC13.** maintain the temperature of steel as per the requirement of downstream process i.e. CCM/VD
- PC14.** collect the final product through tap and transfer it to the other department for further processing

### *Perform post-operation activities*

To be competent, the user/individual on the job must be able to:

- PC15.** update and maintain shift log of production and prepare production and other reports
- PC16.** record all instructions received from higher authorities in the instruction book and all incidents in the occurrence book



## Qualification Pack

**PC17.** inform concerned authority in case of any abnormalities observed during execution of work

### Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** relevant standards and procedures followed in the company
- KU2.** types of ladle furnaces
- KU3.** Standard Operating Procedures (SOP) recommended by OEM for using tools and equipment required during LRF process
- KU4.** basic working process of ladle refining furnace
- KU5.** material required for LRF process
- KU6.** SOP as recommended by competent authority of the organisation
- KU7.** normal running characteristics of LRF equipment
- KU8.** impact of parameters i.e temperature, O<sub>2</sub>, chemical composition etc. on the final product
- KU9.** material like deoxidisers, additives (C, Mn, Si, Al), flux (lime) required during LRF process
- KU10.** need of oxygen and argon gas during LRF process
- KU11.** Desulphurization process of steel
- KU12.** Ca' treatment process for removing inclusions from steel
- KU13.** impact of inclusions on the final product
- KU14.** procedure of operating LRF
- KU15.** how to maintain flow of liquid steel, oxygen, argon during LRF process
- KU16.** need of maintaining temperature of steel during process
- KU17.** post operation activities i.e. inspection, reporting etc.
- KU18.** emergency procedures and safety measures such as gas safety, hot metal handling safety etc. associated with LRF process

### Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read work instructions, equipment manuals and process documents
- GS2.** communicate the process requirements to the supervisor and co-workers
- GS3.** attentively listen and comprehend the information given by the supervisor/team members
- GS4.** write work related information in English/regional language
- GS5.** recognize a workplace problem and take suitable action
- GS6.** plan and organize work according to the work requirements
- GS7.** complete the assigned tasks with minimum supervision
- GS8.** report to the supervisor or deal with a colleague individually, depending on the type of concern



## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                                                                          | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Perform LRF process</i>                                                                                                                                                                | <b>25</b>    | <b>42</b>       | -             | <b>17</b>  |
| <b>PC1.</b> plan the strategy and sequence for adding deoxidizers (i.e. type and quantity), additives and flux in the liquid steel on the basis of steel grade and specification required | 2            | 4               | -             | 2          |
| <b>PC2.</b> add deoxidizers (i.e. type and quantity), additives and flux in the liquid steel as per the planned sequence                                                                  | 2            | 3               | -             | 1          |
| <b>PC3.</b> control the flow of O <sub>2</sub> in the furnace as per the requirement                                                                                                      | 2            | 3               | -             | 1          |
| <b>PC4.</b> heat the liquid steel by using electrodes available in the furnace                                                                                                            | 2            | 3               | -             | 1          |
| <b>PC5.</b> maintain the voltage, current and power in the electrodes as per the requirement                                                                                              | 2            | 3               | -             | 2          |
| <b>PC6.</b> carry out desulphurization process for removing the Sulphur content in the liquid steel by adding more O <sub>2</sub> , lime etc. in it                                       | 3            | 5               | -             | 2          |
| <b>PC7.</b> carry out argon purging in the furnace by opening the nozzle below the ladle during the desulphurization process                                                              | 2            | 4               | -             | 2          |
| <b>PC8.</b> maintain the flow of, pressure and effectiveness of argon gas during purging                                                                                                  | 1            | 2               | -             | 1          |
| <b>PC9.</b> open the tap, at the bottom of ladle and collect the sample of steel to check the Sulphur quantity as per the customer requirement                                            | 2            | 3               | -             | 1          |
| <b>PC10.</b> maintain the temperature of steel as per the requirement                                                                                                                     | 1            | 2               | -             | 1          |
| <b>PC11.</b> add the alloys (ferro alloys) in the steel as per the customer requirement                                                                                                   | 2            | 3               | -             | 1          |
| <b>PC12.</b> clean the steel and remove the inclusions by "Ca" treatment process                                                                                                          | 2            | 3               | -             | 1          |



## Qualification Pack

| Assessment Criteria for Outcomes                                                                                                       | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|----------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC13.</b> maintain the temperature of steel as per the requirement of downstream process i.e. CCM/VD                                | 1            | 2               | -             | 1          |
| <b>PC14.</b> collect the final product through tap and transfer it to the other department for further processing                      | 1            | 2               | -             | -          |
| <i>Perform post-operation activities</i>                                                                                               | <b>5</b>     | <b>8</b>        | -             | <b>3</b>   |
| <b>PC15.</b> update and maintain shift log of production and prepare production and other reports                                      | 2            | 3               | -             | 2          |
| <b>PC16.</b> record all instructions received from higher authorities in the instruction book and all incidents in the occurrence book | 2            | 3               | -             | 1          |
| <b>PC17.</b> inform concerned authority in case of any abnormalities observed during execution of work                                 | 1            | 2               | -             | -          |
| <b>NOS Total</b>                                                                                                                       | <b>30</b>    | <b>50</b>       | -             | <b>20</b>  |





## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                                         |
|----------------------------|---------------------------------------------------------|
| <b>NOS Code</b>            | ISC/N1412                                               |
| <b>NOS Name</b>            | Perform and control Ladle Refining Furnace(LRF) process |
| <b>Sector</b>              | Iron and Steel                                          |
| <b>Sub-Sector</b>          |                                                         |
| <b>Occupation</b>          | Steel Making                                            |
| <b>NSQF Level</b>          | 6                                                       |
| <b>Credits</b>             | 4                                                       |
| <b>Version</b>             | 1.0                                                     |
| <b>Last Reviewed Date</b>  | 30/04/2024                                              |
| <b>Next Review Date</b>    | 30/04/2027                                              |
| <b>NSQC Clearance Date</b> | 30/04/2024                                              |



## Qualification Pack

### ISC/N1704: Apply Decarbonization practice in plant

#### Description

This OS unit is about apply decarbonization practices in steel plant to reduce carbon emissions.

#### Scope

The scope covers the following :

- Apply decarbonization practices

#### Elements and Performance Criteria

##### *Apply decarbonization practices*

To be competent, the user/individual on the job must be able to:

- PC1.** identify various processes (i.e. BOF, DRI, EAF etc.) and fuels (i.e. oil, coal etc.) used in plant during production
- PC2.** calculate carbon emission by summing the emissions resulting from every stage of a product or service's lifetime (material production, manufacturing, use, and end-of-life)
- PC3.** implement efficient technologies and processes in production process that reduce energy use and emissions significantly to achieve net-zero carbon emission in plant
- PC4.** implement technologies such as Carbon capture, usage and storage (CCUS) in production process to reduce and capture carbon emissions in the steelmaking process
- PC5.** replace fossil fuels in BOF, DRI etc. with low/no carbon options such as biofuels, natural gas, hydrogen etc.
- PC6.** use energy sources such as wind, solar, nuclear, and water power combined with advances in electricity storage for minimal CO<sub>2</sub> emissions
- PC7.** use equipment that are powered by electricity instead of fossil fuels
- PC8.** reuse or recycle scrap and waste generated during steelmaking process
- PC9.** switch to renewable electricity and electrifying their vehicle fleets
- PC10.** increase the usage of recycled materials
- PC11.** avoid single use plastics and paper products

#### Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** relevant standards and procedures followed in the company
- KU2.** various fuels such as oil, coal etc. responsible for carbon emission in steel plant
- KU3.** various processes such as BOF, DRI, EAF, rotary kiln etc. responsible for carbon emission in steel plant
- KU4.** gases generated in carbon emission i.e. carbon dioxide (CO<sub>2</sub>), methane (CH<sub>4</sub>), and nitrous oxide (N<sub>2</sub>O) etc.
- KU5.** CO<sub>2</sub> emission and its effect on environment



## Qualification Pack

- KU6.** Five R's refuse, reduce, reuse, rot, recycle
- KU7.** Carbon capture, usage and storage (CCUS) technology
- KU8.** carbon dioxide removal (CDR) or negative emissions techniques (NETs)
- KU9.** how to calculate CO2 using different fuels
- KU10.** how to control burning of fuel
- KU11.** renewable energy sources such as wind, solar, nuclear, and water

## Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read work instructions, equipment manuals and process documents
- GS2.** communicate the process requirements to the supervisor and co-workers
- GS3.** attentively listen and comprehend the information given by the supervisor/team members
- GS4.** write work related information in English/regional language
- GS5.** recognize a workplace problem and take suitable action
- GS6.** plan and organize work according to the work requirements
- GS7.** complete the assigned tasks with minimum supervision
- GS8.** report to the supervisor or deal with a colleague individually, depending on the type of concern



## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                                                                        | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Apply decarbonization practices</i>                                                                                                                                                  | <b>30</b>    | <b>50</b>       | -             | <b>20</b>  |
| <b>PC1.</b> identify various processes (i.e. BOF, DRI, EAF etc.) and fuels (i.e. oil, coal etc.) used in plant during production                                                        | 3            | 5               | -             | 2          |
| <b>PC2.</b> calculate carbon emission by summing the emissions resulting from every stage of a product or service's lifetime (material production, manufacturing, use, and end-of-life) | 3            | 5               | -             | 2          |
| <b>PC3.</b> implement efficient technologies and processes in production process that reduce energy use and emissions significantly to achieve net-zero carbon emission in plant        | 3            | 5               | -             | 2          |
| <b>PC4.</b> implement technologies such as Carbon capture, usage and storage (CCUS) in production process to reduce and capture carbon emissions in the steelmaking process             | 3            | 5               | -             | 2          |
| <b>PC5.</b> replace fossil fuels in BOF, DRI etc. with low/no carbon options such as biofuels, natural gas, hydrogen etc.                                                               | 3            | 5               | -             | 2          |
| <b>PC6.</b> use energy sources such as wind, solar, nuclear, and water power combined with advances in electricity storage for minimal CO2 emissions                                    | 3            | 5               | -             | 2          |
| <b>PC7.</b> use equipment that are powered by electricity instead of fossil fuels                                                                                                       | 3            | 5               | -             | 2          |
| <b>PC8.</b> reuse or recycle scrap and waste generated during steelmaking process                                                                                                       | 3            | 5               | -             | 2          |
| <b>PC9.</b> switch to renewable electricity and electrifying their vehicle fleets                                                                                                       | 2            | 4               | -             | 2          |
| <b>PC10.</b> increase the usage of recycled materials                                                                                                                                   | 2            | 3               | -             | 1          |
| <b>PC11.</b> avoid single use plastics and paper products                                                                                                                               | 2            | 3               | -             | 1          |
| <b>NOS Total</b>                                                                                                                                                                        | <b>30</b>    | <b>50</b>       | -             | <b>20</b>  |



## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                         |
|----------------------------|-----------------------------------------|
| <b>NOS Code</b>            | ISC/N1704                               |
| <b>NOS Name</b>            | Apply Decarbonization practice in plant |
| <b>Sector</b>              | Iron and Steel                          |
| <b>Sub-Sector</b>          |                                         |
| <b>Occupation</b>          | Generic                                 |
| <b>NSQF Level</b>          | 6                                       |
| <b>Credits</b>             | 1                                       |
| <b>Version</b>             | 1.0                                     |
| <b>Last Reviewed Date</b>  | 07/10/2025                              |
| <b>Next Review Date</b>    | 07/10/2028                              |
| <b>NSQC Clearance Date</b> | 07/10/2025                              |



## Qualification Pack

# ISC/N1703: Apply statistical process control (SPC) techniques

## Description

This OS unit is about apply statistical process control (SPC) technique to improve the product quality and production efficiency.

## Scope

The scope covers the following :

- Apply statistical process control (SPC) technique

## Elements and Performance Criteria

### *Apply statistical process control (SPC) techniques*

To be competent, the user/individual on the job must be able to:

- PC1.** identify the processes that impacts the output of the product
- PC2.** identify the attributes that need to be measured during production
- PC3.** develop a subgroup strategy and sampling plan based on the product's criticality
- PC4.** collect the data per sample size and select an appropriate SPC chart based on data type (Continuous or Discrete) and subgroup size
- PC5.** calculate standard deviation and variance of the production
- PC6.** prepare normal distribution curve on the basis of data collected
- PC7.** calculate the upper control limit (UCL) and lower control limit (LCL) to describe the natural variation of attributes
- PC8.** calculate process capabilities by applying standard formula
- PC9.** calculate process capability index and draw graph on the basis of it
- PC10.** identify improvements required in process on the basis of observations of process capability index results
- PC11.** monitor process variation by interpreting the control chart and check whether any point is out of control and the pattern

## Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** six sigma concept
- KU2.** statistical process control (SPC) technique and its importance
- KU3.** SPC concept i.e. mode, mean, median, mad
- KU4.** benefits of SPC
- KU5.** objectives of SPC
- KU6.** various tools of SPC
- KU7.** how to perform SPC



## Qualification Pack

- KU8.** standard deviation and variance
- KU9.** how to prepare normal distribution curve
- KU10.** Process capability index and its calculation formula
- KU11.** how to prepare control charts
- KU12.** concept of accuracy and precision

## Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read work instructions and process documents
- GS2.** communicate the process requirements to the supervisor and co-workers
- GS3.** attentively listen and comprehend the information given by the supervisor/team members
- GS4.** write work related information in English/regional language
- GS5.** recognize a workplace problem and take suitable action
- GS6.** plan and organize work according to the work requirements
- GS7.** complete the assigned tasks with minimum supervision
- GS8.** report to the supervisor or deal with a colleague individually, depending on the type of concern



## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                               | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Apply statistical process control (SPC) techniques</i>                                                                                      | <b>30</b>    | <b>50</b>       | -             | <b>20</b>  |
| <b>PC1.</b> identify the processes that impacts the output of the product                                                                      | 2            | 3               | -             | 1          |
| <b>PC2.</b> identify the attributes that need to be measured during production                                                                 | 2            | 3               | -             | 1          |
| <b>PC3.</b> develop a subgroup strategy and sampling plan based on the product's criticality                                                   | 3            | 5               | -             | 2          |
| <b>PC4.</b> collect the data per sample size and select an appropriate SPC chart based on data type (Continuous or Discrete) and subgroup size | 3            | 5               | -             | 2          |
| <b>PC5.</b> calculate standard deviation and variance of the production                                                                        | 3            | 5               | -             | 2          |
| <b>PC6.</b> prepare normal distribution curve on the basis of data collected                                                                   | 3            | 5               | -             | 2          |
| <b>PC7.</b> calculate the upper control limit (UCL) and lower control limit (LCL) to describe the natural variation of attributes              | 3            | 5               | -             | 2          |
| <b>PC8.</b> calculate process capabilities by applying standard formula                                                                        | 3            | 5               | -             | 2          |
| <b>PC9.</b> calculate process capability index and draw graph on the basis of it                                                               | 3            | 5               | -             | 2          |
| <b>PC10.</b> identify improvements required in process on the basis of observations of process capability index results                        | 2            | 4               | -             | 2          |
| <b>PC11.</b> monitor process variation by interpreting the control chart and check whether any point is out of control and the pattern         | 3            | 5               | -             | 2          |
| <b>NOS Total</b>                                                                                                                               | <b>30</b>    | <b>50</b>       | -             | <b>20</b>  |





## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                                    |
|----------------------------|----------------------------------------------------|
| <b>NOS Code</b>            | ISC/N1703                                          |
| <b>NOS Name</b>            | Apply statistical process control (SPC) techniques |
| <b>Sector</b>              | Iron and Steel                                     |
| <b>Sub-Sector</b>          |                                                    |
| <b>Occupation</b>          | Generic                                            |
| <b>NSQF Level</b>          | 6                                                  |
| <b>Credits</b>             | 2                                                  |
| <b>Version</b>             | 1.0                                                |
| <b>Last Reviewed Date</b>  | 07/10/2025                                         |
| <b>Next Review Date</b>    | 07/10/2028                                         |
| <b>NSQC Clearance Date</b> | 07/10/2025                                         |



## Qualification Pack

### DGT/VSQ/N0103: Employability Skills (90 Hours)

#### Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

#### Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

#### Elements and Performance Criteria

##### *Introduction to Employability Skills*

To be competent, the user/individual on the job must be able to:

- PC1.** understand the significance of employability skills in meeting the current job market requirement and future of work
- PC2.** identify and explore learning and employability relevant portals
- PC3.** research about the different industries, job market trends, latest skills required and the available opportunities

##### *Constitutional values – Citizenship*

To be competent, the user/individual on the job must be able to:

- PC4.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC5.** follow environmentally sustainable practices

##### *Becoming a Professional in the 21st Century*

To be competent, the user/individual on the job must be able to:

- PC6.** recognize the significance of 21st Century Skills for employment



## Qualification Pack

- PC7.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life
- PC8.** adopt a continuous learning mindset for personal and professional development

### *Basic English Skills*

To be competent, the user/individual on the job must be able to:

- PC9.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC10.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC11.** write short messages, notes, letters, e-mails etc. in English

### *Career Development & Goal Setting*

To be competent, the user/individual on the job must be able to:

- PC12.** identify career goals based on the skills, interests, knowledge, and personal attributes
- PC13.** prepare a career development plan with short- and long-term goals

### *Communication Skills*

To be competent, the user/individual on the job must be able to:

- PC14.** follow verbal and non-verbal communication etiquette while communicating in professional and public settings
- PC15.** use active listening techniques for effective communication
- PC16.** communicate in writing using appropriate style and format based on formal or informal requirements
- PC17.** work collaboratively with others in a team

### *Diversity & Inclusion*

To be competent, the user/individual on the job must be able to:

- PC18.** communicate and behave appropriately with all genders and PwD
- PC19.** escalate any issues related to sexual harassment at workplace according to POSH Act

### *Financial and Legal Literacy*

To be competent, the user/individual on the job must be able to:

- PC20.** identify and select reliable institutions for various financial products and services such as bank account, debit and credit cards, loans, insurance etc.
- PC21.** carry out offline and online financial transactions, safely and securely, using various methods and check the entries in the passbook
- PC22.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC23.** identify relevant rights and laws and use legal aids to fight against legal exploitation

### *Essential Digital Skills*

To be competent, the user/individual on the job must be able to:

- PC24.** operate digital devices and use their features and applications securely and safely
- PC25.** carry out basic internet operations by connecting to the internet safely and securely, using the mobile data or other available networks through Bluetooth, Wi-Fi, etc.
- PC26.** display responsible online behaviour while using various social media platforms



## Qualification Pack

- PC27.** create a personal email account, send and process received messages as per requirement
- PC28.** carry out basic procedures in documents, spreadsheets and presentations using respective and appropriate applications
- PC29.** utilize virtual collaboration tools to work effectively

### *Entrepreneurship*

To be competent, the user/individual on the job must be able to:

- PC30.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC31.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC32.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

### *Customer Service*

To be competent, the user/individual on the job must be able to:

- PC33.** identify different types of customers and ways to communicate with them
- PC34.** identify and respond to customer requests and needs in a professional manner
- PC35.** use appropriate tools to collect customer feedback
- PC36.** follow appropriate hygiene and grooming standards

### *Getting ready for apprenticeship & Jobs*

To be competent, the user/individual on the job must be able to:

- PC37.** create a professional Curriculum vitae (Résumé)
- PC38.** search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively
- PC39.** apply to identified job openings using offline /online methods as per requirement
- PC40.** answer questions politely, with clarity and confidence, during recruitment and selection
- PC41.** identify apprenticeship opportunities and register for it as per guidelines and requirements

## Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** need for employability skills and different learning and employability related portals
- KU2.** various constitutional and personal values
- KU3.** different environmentally sustainable practices and their importance
- KU4.** Twenty first (21st) century skills and their importance
- KU5.** how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up
- KU6.** importance of career development and setting long- and short-term goals
- KU7.** about effective communication
- KU8.** POSH Act
- KU9.** Gender sensitivity and inclusivity
- KU10.** different types of financial institutes, products, and services



## Qualification Pack

- KU11.** components of salary and how to compute income and expenditure
- KU12.** importance of maintaining safety and security in offline and online financial transactions
- KU13.** different legal rights and laws
- KU14.** different types of digital devices and the procedure to operate them safely and securely
- KU15.** how to create and operate an e- mail account
- KU16.** use applications such as word processors, spreadsheets etc.
- KU17.** how to identify business opportunities
- KU18.** types and needs of customers
- KU19.** how to apply for a job and prepare for an interview
- KU20.** apprenticeship scheme and the process of registering on apprenticeship portal

## Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read and write different types of documents/instructions/correspondence in English and other languages
- GS2.** communicate effectively using appropriate language in formal and informal settings
- GS3.** behave politely and appropriately with all to maintain effective work relationship
- GS4.** how to work in a virtual mode, using various technological platforms
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection



## Qualification Pack

### Assessment Criteria

| Assessment Criteria for Outcomes                                                                                                                                                                                                                                                                                 | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <i>Introduction to Employability Skills</i>                                                                                                                                                                                                                                                                      | <b>1</b>     | <b>1</b>        | -             | -          |
| <b>PC1.</b> understand the significance of employability skills in meeting the current job market requirement and future of work                                                                                                                                                                                 | -            | -               | -             | -          |
| <b>PC2.</b> identify and explore learning and employability relevant portals                                                                                                                                                                                                                                     | -            | -               | -             | -          |
| <b>PC3.</b> research about the different industries, job market trends, latest skills required and the available opportunities                                                                                                                                                                                   | -            | -               | -             | -          |
| <i>Constitutional values – Citizenship</i>                                                                                                                                                                                                                                                                       | <b>1</b>     | <b>1</b>        | -             | -          |
| <b>PC4.</b> recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.                                                               | -            | -               | -             | -          |
| <b>PC5.</b> follow environmentally sustainable practices                                                                                                                                                                                                                                                         | -            | -               | -             | -          |
| <i>Becoming a Professional in the 21st Century</i>                                                                                                                                                                                                                                                               | <b>1</b>     | <b>3</b>        | -             | -          |
| <b>PC6.</b> recognize the significance of 21st Century Skills for employment                                                                                                                                                                                                                                     | -            | -               | -             | -          |
| <b>PC7.</b> practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life | -            | -               | -             | -          |
| <b>PC8.</b> adopt a continuous learning mindset for personal and professional development                                                                                                                                                                                                                        | -            | -               | -             | -          |
| <i>Basic English Skills</i>                                                                                                                                                                                                                                                                                      | <b>3</b>     | <b>4</b>        | -             | -          |
| <b>PC9.</b> use basic English for everyday conversation in different contexts, in person and over the telephone                                                                                                                                                                                                  | -            | -               | -             | -          |



## Qualification Pack

| Assessment Criteria for Outcomes                                                                                                                                       | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC10.</b> read and understand routine information, notes, instructions, mails, letters etc. written in English                                                      | -            | -               | -             | -          |
| <b>PC11.</b> write short messages, notes, letters, e-mails etc. in English                                                                                             | -            | -               | -             | -          |
| <i>Career Development &amp; Goal Setting</i>                                                                                                                           | <b>1</b>     | <b>2</b>        | -             | -          |
| <b>PC12.</b> identify career goals based on the skills, interests, knowledge, and personal attributes                                                                  | -            | -               | -             | -          |
| <b>PC13.</b> prepare a career development plan with short- and long-term goals                                                                                         | -            | -               | -             | -          |
| <i>Communication Skills</i>                                                                                                                                            | <b>2</b>     | <b>2</b>        | -             | -          |
| <b>PC14.</b> follow verbal and non-verbal communication etiquette while communicating in professional and public settings                                              | -            | -               | -             | -          |
| <b>PC15.</b> use active listening techniques for effective communication                                                                                               | -            | -               | -             | -          |
| <b>PC16.</b> communicate in writing using appropriate style and format based on formal or informal requirements                                                        | -            | -               | -             | -          |
| <b>PC17.</b> work collaboratively with others in a team                                                                                                                | -            | -               | -             | -          |
| <i>Diversity &amp; Inclusion</i>                                                                                                                                       | <b>1</b>     | <b>1</b>        | -             | -          |
| <b>PC18.</b> communicate and behave appropriately with all genders and PwD                                                                                             | -            | -               | -             | -          |
| <b>PC19.</b> escalate any issues related to sexual harassment at workplace according to POSH Act                                                                       | -            | -               | -             | -          |
| <i>Financial and Legal Literacy</i>                                                                                                                                    | <b>2</b>     | <b>3</b>        | -             | -          |
| <b>PC20.</b> identify and select reliable institutions for various financial products and services such as bank account, debit and credit cards, loans, insurance etc. | -            | -               | -             | -          |
| <b>PC21.</b> carry out offline and online financial transactions, safely and securely, using various methods and check the entries in the passbook                     | -            | -               | -             | -          |



## Qualification Pack

| Assessment Criteria for Outcomes                                                                                                                                                     | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC22.</b> identify common components of salary and compute income, expenses, taxes, investments etc                                                                               | -            | -               | -             | -          |
| <b>PC23.</b> identify relevant rights and laws and use legal aids to fight against legal exploitation                                                                                | -            | -               | -             | -          |
| <i>Essential Digital Skills</i>                                                                                                                                                      | <b>3</b>     | <b>5</b>        | -             | -          |
| <b>PC24.</b> operate digital devices and use their features and applications securely and safely                                                                                     | -            | -               | -             | -          |
| <b>PC25.</b> carry out basic internet operations by connecting to the internet safely and securely, using the mobile data or other available networks through Bluetooth, Wi-Fi, etc. | -            | -               | -             | -          |
| <b>PC26.</b> display responsible online behaviour while using various social media platforms                                                                                         | -            | -               | -             | -          |
| <b>PC27.</b> create a personal email account, send and process received messages as per requirement                                                                                  | -            | -               | -             | -          |
| <b>PC28.</b> carry out basic procedures in documents, spreadsheets and presentations using respective and appropriate applications                                                   | -            | -               | -             | -          |
| <b>PC29.</b> utilize virtual collaboration tools to work effectively                                                                                                                 | -            | -               | -             | -          |
| <i>Entrepreneurship</i>                                                                                                                                                              | <b>2</b>     | <b>3</b>        | -             | -          |
| <b>PC30.</b> identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research                                           | -            | -               | -             | -          |
| <b>PC31.</b> develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion                                                          | -            | -               | -             | -          |
| <b>PC32.</b> identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity                                               | -            | -               | -             | -          |
| <i>Customer Service</i>                                                                                                                                                              | <b>1</b>     | <b>2</b>        | -             | -          |
| <b>PC33.</b> identify different types of customers and ways to communicate with them                                                                                                 | -            | -               | -             | -          |





## Qualification Pack

| Assessment Criteria for Outcomes                                                                                                                                                 | Theory Marks | Practical Marks | Project Marks | Viva Marks |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|
| <b>PC34.</b> identify and respond to customer requests and needs in a professional manner                                                                                        | -            | -               | -             | -          |
| <b>PC35.</b> use appropriate tools to collect customer feedback                                                                                                                  | -            | -               | -             | -          |
| <b>PC36.</b> follow appropriate hygiene and grooming standards                                                                                                                   | -            | -               | -             | -          |
| <i>Getting ready for apprenticeship &amp; Jobs</i>                                                                                                                               | <b>2</b>     | <b>3</b>        | -             | -          |
| <b>PC37.</b> create a professional Curriculum vitae (Résumé)                                                                                                                     | -            | -               | -             | -          |
| <b>PC38.</b> search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively | -            | -               | -             | -          |
| <b>PC39.</b> apply to identified job openings using offline /online methods as per requirement                                                                                   | -            | -               | -             | -          |
| <b>PC40.</b> answer questions politely, with clarity and confidence, during recruitment and selection                                                                            | -            | -               | -             | -          |
| <b>PC41.</b> identify apprenticeship opportunities and register for it as per guidelines and requirements                                                                        | -            | -               | -             | -          |
| <b>NOS Total</b>                                                                                                                                                                 | <b>20</b>    | <b>30</b>       | -             | -          |



## Qualification Pack

### National Occupational Standards (NOS) Parameters

|                            |                                 |
|----------------------------|---------------------------------|
| <b>NOS Code</b>            | DGT/VSQ/N0103                   |
| <b>NOS Name</b>            | Employability Skills (90 Hours) |
| <b>Sector</b>              | Cross Sectoral                  |
| <b>Sub-Sector</b>          | Professional Skills             |
| <b>Occupation</b>          | Employability                   |
| <b>NSQF Level</b>          | 5                               |
| <b>Credits</b>             | 3                               |
| <b>Version</b>             | 1.0                             |
| <b>Last Reviewed Date</b>  | 07/10/2025                      |
| <b>Next Review Date</b>    | 07/10/2028                      |
| <b>NSQC Clearance Date</b> | 07/10/2025                      |

### Assessment Guidelines and Assessment Weightage

#### Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Performance Criteria (PC) (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.
3. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).
4. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.
5. In case of successfully passing only certain number of NOSs, the trainee is eligible to take subsequent assessment on the balance NOS's to pass the Qualification Pack.
6. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack

#### Minimum Aggregate Passing % at QP Level : 70



## Qualification Pack

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

### Assessment Weightage

Compulsory NOS

| National Occupational Standards                                              | Theory Marks | Practical Marks | Project Marks | Viva Marks | Total Marks | Weightage  |
|------------------------------------------------------------------------------|--------------|-----------------|---------------|------------|-------------|------------|
| ISC/N0946.Manage and implement health and safety practices at the work place | 30           | 50              | -             | 20         | 100         | 10         |
| ISC/N0947.Manage team and others at workplace                                | 30           | 50              | -             | 20         | 100         | 10         |
| ISC/N1407.Prepare for Ladle Refining Furnace (LRF) Process                   | 30           | 50              | -             | 20         | 100         | 20         |
| ISC/N1412.Perform and control Ladle Refining Furnace(LRF) process            | 30           | 50              | -             | 20         | 100         | 30         |
| ISC/N1704.Apply Decarbonization practice in plant                            | 30           | 50              | 0             | 20         | 100         | 10         |
| ISC/N1703.Apply statistical process control (SPC) techniques                 | 30           | 50              | 0             | 20         | 100         | 10         |
| DGT/VSQ/N0103.Employability Skills (90 Hours)                                | 20           | 30              | -             | -          | 50          | 10         |
| <b>Total</b>                                                                 | <b>200</b>   | <b>330</b>      | <b>0</b>      | <b>120</b> | <b>650</b>  | <b>100</b> |



## Qualification Pack

### Acronyms

|             |                                                 |
|-------------|-------------------------------------------------|
| <b>NOS</b>  | National Occupational Standard(s)               |
| <b>NSQF</b> | National Skills Qualifications Framework        |
| <b>QP</b>   | Qualifications Pack                             |
| <b>TVET</b> | Technical and Vocational Education and Training |



## Qualification Pack

### Glossary

|                                              |                                                                                                                                                                                                                                                                                                  |
|----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Sector</b>                                | Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.                                                                      |
| <b>Sub-sector</b>                            | Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.                                                                                                                                                                                     |
| <b>Occupation</b>                            | Occupation is a set of job roles, which perform similar/ related set of functions in an industry.                                                                                                                                                                                                |
| <b>Job role</b>                              | Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.                                                                                                                                                                                |
| <b>Occupational Standards (OS)</b>           | OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts. |
| <b>Performance Criteria (PC)</b>             | Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.                                                                                                                                                                    |
| <b>National Occupational Standards (NOS)</b> | NOS are occupational standards which apply uniquely in the Indian context.                                                                                                                                                                                                                       |
| <b>Qualifications Pack (QP)</b>              | QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.                                                                                                                       |
| <b>Unit Code</b>                             | Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'                                                                                                                                                                                                        |
| <b>Unit Title</b>                            | Unit title gives a clear overall statement about what the incumbent should be able to do.                                                                                                                                                                                                        |
| <b>Description</b>                           | Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.                                                                                                                   |
| <b>Scope</b>                                 | Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.                                                                                            |



## Qualification Pack

|                                         |                                                                                                                                                                                                                                                                                                                                                                        |
|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Knowledge and Understanding (KU)</b> | Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.                                                                                                                                               |
| <b>Organisational Context</b>           | Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.                                                                                                                                                                       |
| <b>Technical Knowledge</b>              | Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.                                                                                                                                                                                                                                                               |
| <b>Core Skills/ Generic Skills (GS)</b> | Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles. |
| <b>Electives</b>                        | Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.                                                                            |
| <b>Options</b>                          | Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.                                                                                                                                                          |